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A multi-technique investigation of sulfur grain boundary segregation in nickel

M. Allart,[a] F. Christien,[a,*] R. Le Gall,[a] P. Nowakowski[a] and C.R.M. Grovenor[b]

[a] *Institut des Matériaux Jean Rouxel (IMN), Université de Nantes, CNRS, Rue Christian Pauc, BP 50609, 44306 Nantes Cedex 3, France*

[b] *Department of Materials, University of Oxford, Parks Road, Oxford, UK*

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The grain boundary segregation (GBS) of sulfur in nickel was studied using Auger electron spectroscopy (AES) and wavelength-dispersive X-ray spectroscopy (WDS), and a linear correlation between the two techniques was demonstrated. The kinetics and thermodynamic parameters of sulfur GBS were determined. The discrepancy between the AES/WDS results obtained here and those from a previous study on the same material undertaken using secondary ion mass spectroscopy was interpreted in terms of the selection of grain boundary type by the sample preparation method.
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The detrimental effect of sulfur grain boundary segregation (GBS) on the mechanical properties of nickel and nickel alloys has been documented in several experimental studies [1,2]. The temperature dependence of equilibrium solute GBS obeys the well-known Langmuir–McLean equation [3]:

$$\frac{X}{X_{Max}-X}=\frac{C_V}{1-C_V}\exp\left(-\frac{\Delta G}{RT}\right), \quad (1)$$

where X is the solute grain boundary (GB) concentration, X_{Max} is the solute grain boundary concentration at saturation, C_V is the bulk solute concentration, ΔG is the grain boundary segregation free energy (J mol^{-1}), R is the gas constant (8.314 J mol^{-1} K^{-1}) and T is the temperature (K). In Eq. (1), X can be expressed as a fraction of a monolayer or as a mass (or a number of atoms) of solute per unit grain boundary area. Only two experimental studies have been dedicated to the determination of the thermodynamic parameters (ΔG and X_{Max}) of sulfur GBS [4,5]. However, the values published in these studies must be considered with caution. The reliability of the ΔG value proposed by Larère et al. [4] can be seriously questioned since it was obtained "by supposing that at 700 °C, the maximum equilibrium segregation is at least 95% of the saturation value". However, no measurement of sulfur GBS supporting that assumption is presented in that work. Mulford et al. [5] have determined $\Delta G = 72.8$ kJ mol^{-1} from Auger measurements undertaken on specimens equilibrated at several different temperatures in the range 500–900 °C. However, they used nickel containing a bulk concentration of sulfur $C_V = 75$ at. ppm, which is very likely to be above the limit of solubility, at least in the temperature range 500–700 °C [6]. This means that the term C_V is not accurately known in that study.

The kinetics of GBS during isothermal annealing obeys the McLean kinetic equation [3]:

$$X = X_0 + \frac{4C_V\sqrt{Dt}}{\sqrt{\pi}}, \quad (2)$$

where X_0 is the solute grain boundary concentration before the annealing treatment to induce segregation, D is the solute bulk diffusion coefficient (cm^2 s^{-1}) at the annealing temperature and t is the annealing time. In Eq. (2), the bulk solute concentration, C_V, is expressed in cm^{-3} or g cm^{-3} (number of atoms or mass of solute per unit volume) and the GB concentration, X, is in cm^{-2} or g cm^{-2}. The sulfur bulk diffusion coefficient has been determined by Vladimirov et al. [7] using a radiotracer depth profiling technique on Ni crystals in the temperature range 800–1200 °C.

* Corresponding author. Tel.: +33 2 40 68 31 72; fax: +33 2 40 68 31 99; e-mail: frederic.christien@univ-nantes.fr

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For the last 50 years, Auger electron spectroscopy (AES) has been the most widely used technique for the quantification of GBS [8,9]. The analysis is undertaken on an intergranular fracture surface, and since AES is highly sensitive to surface contamination, the specimen has to be fractured in situ, i.e. inside the ultrahigh-vacuum (UHV) chamber of the spectrometer. GBS measured using AES is expressed as a fraction of a monolayer [8,9]. Over the last five years, new experimental techniques have been proposed for the quantitative analysis of GBS. Nowakowski et al. [10] have demonstrated the possibility of quantitative determination of GBS using wavelength-dispersive X-ray spectroscopy (WDS) in a scanning electron microscope (SEM). Like AES, WDS analysis is undertaken on an intergranular fracture surface. However, in contrast to AES, the specimen can be fractured ex situ, as it has been shown that WDS is practically insensitive to surface contamination. The WDS technique is, however, limited to systems with very low solute concentrations in the bulk. GBS as measured using WDS is expressed as a mass (or a number of atoms) of solute per unit surface in the grain boundary.

The possibility of quantitative determination of GBS by secondary ion mass spectroscopy (SIMS) using a NanoSIMS system has also been recently demonstrated by Christien et al. [11,12]. The NanoSIMS ion microprobe uses a highly focused (~50 nm) Cs^+ primary ion beam and a high-resolution magnetic sector mass spectrometer. The analysis of GBS using NanoSIMS is undertaken on a polished cross-section of bulk material, in contrast to AES and WDS where the analysis is done on a fracture surface. This is a major advantage as it allows the analysis of all the grain boundaries that intersect the polished surface, not only those that can be fractured as is the case for AES and WDS. As the lateral resolution of NanoSIMS is far above the GB width, the beam broadening effect is taken into account in the quantification procedure [11,12]. As for WDS, GBS measured using NanoSIMS is expressed as a mass (or a number of atoms) of solute per unit area of grain boundary.

The objective of this paper is the determination of the kinetics and thermodynamics of sulfur GBS using AES and WDS. A comparison with the data obtained in previous studies [11,12] using NanoSIMS on the same material is then presented.

The material used in this study is ultrahigh-purity nickel from Aperam. A 6 kg cast was prepared with controlled additions of sulfur. A sulfur bulk concentration of 5.4 wt. ppm was measured using glow discharge mass spectroscopy. The material was provided as 3 mm thick plates, annealed at 900 °C. The average grain size was 260 μm.

AES measurements were undertaken using a MAC2 cylindrical spectrometer from Riber. Each specimen analyzed was first fractured at approximately −120 °C using a tensile apparatus inside the UHV preparation chamber (~5×10^{-10} mbar) of the spectrometer. One half of the broken specimen was then transferred to the analysis chamber (~10^{-9} mbar). About 30 GB facets were analyzed for each specimen. Sulfur was the only segregated element detected at grain boundaries. Sulfur GBS is quantified using the Seah equation [8]:

$$X = \frac{1}{1 - \exp\left(-\frac{a}{\lambda \cos\beta}\right)} \frac{\frac{H_S}{S_S}}{\frac{H_S}{S_S} + \frac{H_{Ni}}{S_{Ni}}}, \quad (3)$$

where X is the sulfur GB concentration expressed as a fraction of a monolayer, λ is the sulfur Auger electron attenuation length, a is the thickness of the segregation layer ($\lambda \approx 2a$ in our case [13]), β is the average emission angle ($\beta = 45°$ in our case), H_S and H_{Ni} are the sulfur and nickel peak heights measured on the derivative spectrum at 152 and 848 eV, respectively, and S_S and S_{Ni} are the sulfur and nickel sensitivity factors, respectively. Following the approach described in Ref. [14], S_S and S_{Ni} have been measured on our spectrometer using FeS_2 and FeNi 36% (Invar) standard materials, respectively, giving $S_S = 13.76$ and $S_{Ni} = 1.53$. The standard deviation of the sulfur concentration measured on different GB facets of the same specimen was about 35% of the mean value. There are several possible contributions to this significant variation. The first is that the sulfur equilibrium GB concentration depends on the grain boundary character and is consequently different from one grain boundary to another. The second possible contribution is the fact the sulfur is split randomly (not equally) on both sides of the fracture surface. Furthermore, it has been demonstrated in Ref. [15] that another very important contribution is the variation of the emission angle from one GB facet to another (at least in the spectrometer used in this study), which was not taken into account for each individual GB facet. Even with these experimental perturbations, the overall sulfur GBS in the specimen can be determined accurately by replicating the analysis on a large number of GB facets (here, about 30 for each specimen) and calculating the average value. Using this methodology, the statistical uncertainty (repeatability) on the overall sulfur GBS measured in one particular specimen using AES is reduced to less than 10%, which corresponds to a maximum inaccuracy of a few per cent of a monolayer.

WDS measurements were undertaken using an Inca Wave spectrometer from Oxford Instruments in a high beam current Merlin SEM from Carl Zeiss. Each specimen to be analyzed was fractured ex situ on a conventional tensile machine in a liquid nitrogen container. One half of the broken specimen was then transferred into the SEM chamber (~10^{-6} mbar) for WDS analysis. 15 GB facets on average were analyzed for each specimen using a beam voltage of 20 kV and beam currents ranging from 350 to 450 nA. The sulfur GBS was quantified using the following equation [10]:

$$X = 2K \frac{I}{I_{Std}} \cos\theta - X_V, \quad (4)$$

where X is the sulfur GB concentration expressed as a mass of sulfur per unit surface (ng cm^{-2}), K is a constant taking account of the element analyzed (sulfur), the chemical composition of the substrate (pure nickel), the X-ray line chosen for analysis (K_α), the beam voltage used (20 kV) and the composition of the standard material (FeS_2). The determination of K for this system (256100 ng cm^{-2}) is detailed in Ref. [10]. I and I_{Std} (counts s^{-1} nA^{-1}) are the intensities of the sulfur K_α line measured on the grain boundary facets of the specimen

and on the standard material (FeS_2), respectively. θ is the tilt angle of the GB facet analyzed (the angle between the facet normal and the primary beam). For each facet analyzed, the tilt angle θ is determined from the absorbed specimen current (measured using a pico-ammeter) following the methodology described in Ref. [10]. X_V is a correction term taking account of the sulfur signal coming from the bulk sample. With the analysis conditions used here, this term was estimated at 4.0 ng cm^{-2} using the approach described in Ref. [10].

The standard deviation of the sulfur concentration measured on different GB facets of the same specimen was about 30% of the mean value for similar reasons as described above for AES (sulfur on both sides of the fracture surface and the dependence of the equilibrium sulfur GB concentration on the grain boundary character). With about 15 GB facets analyzed, the statistical uncertainty (repeatability) on the overall sulfur GBS measured in one particular specimen using WDS is about 10%, again corresponding to a maximum inaccuracy of few per cent of a monolayer.

In this work, 22 specimens have been equilibrated at four different temperatures (550, 650, 750 and 900 °C) and analyzed either by AES or WDS. The minimum amount of time needed to achieve equilibration at a given temperature was determined from Eq. (2). It varies from 30 min at 900 °C to several months at 550 °C. For each temperature, two to five different equilibrated specimens were analyzed using each technique.

Figure 1 shows the correlation between the GB sulfur concentrations measured using WDS and AES, which can be fitted by the following equation:

$$X_{WDS} = 67.6 X_{AES}, \quad (5)$$

where X_{AES} is expressed as a fraction of a sulfur monolayer and X_{WDS} is expressed as the mass of sulfur per unit grain boundary area in ng cm^{-2}. Using Eq. (5), it is concluded that a full sulfur monolayer in the grain boundary would correspond to 67.6 ng cm^{-2} of sulfur, i.e. 1.3×10^{15} sulfur atoms per cm^2. This planar density is intermediate between the atomic density on a (100) and a (110) nickel plane (1.61×10^{15} and 1.14×10^{15} cm^{-2}, respectively). The deviation of the experimental points in Figure 1 from a linear fit can be accounted for by slightly different sulfur bulk contents in the specimens used, resulting in different levels of equilibrium segregation at a given temperature. The heterogeneity of the sulfur bulk content is in fact about 50% in the material used in this study, as demonstrated in Ref. [11] using SIMS.

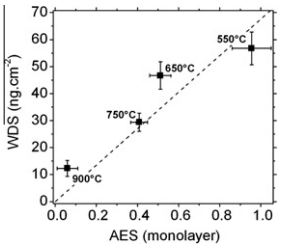


Figure 1. Correlation between the GB sulfur concentration measured using AES and WDS on specimens equilibrated at 550, 650, 750 and 900 °C. Each point corresponds to the average value from two to five specimens analyzed using each technique.

The kinetics of sulfur GBS was investigated at 550 and 750 °C, using specimens with different annealing times, and the results are shown in Figure 2. In the initial state (the material annealed at 900 °C), the average sulfur GB concentration is 12 ng cm^{-2}. In the transitory regime, i.e. before equilibrium is reached, the GB sulfur concentration is proportional to the square root of time. The linear fit of the experimental data in this regime allows the determination of the sulfur bulk diffusion coefficient, according to Eq. (2). The values obtained ($D_{550°C} = 2.8 \times 10^{-14}$ cm^2 s^{-1} and $D_{750°C} = 1.7 \times 10^{-12}$ cm^2 s^{-1}) are shown in an Arrhenius plot in Figure 3. They are in good agreement with the diffusion coefficients determined by Vladimirov et al. at higher temperature (800–1200 °C) [7].

Figure 4 shows the temperature dependence of the equilibrium GB sulfur concentration determined using AES and WDS. The AES and WDS data were fitted using Eq. (1). The segregation entropy ΔS was first neglected. The fit procedure then gives values of segregation enthalpy $\Delta H = -93.9 \pm 2.5$ kJ mol^{-1} and GB sulfur concentration at saturation $X_{Max} = 0.97 \pm 0.09$ monolayer $= 65.8 \pm 6.1$ ng cm^{-2}. The fit procedure was also replicated using the usual expression of the segregation free energy ($\Delta G = \Delta H + T\Delta S$) [9,16]. However, the fit was not significantly improved and a huge error (±150%) in the GB segregation entropy ΔS was obtained. Consequently, considering the segregation entropy was not deemed of particular interest here. It should be noted that X_{Max}, the sulfur GB concentration at saturation, measured here (65.8 ng cm^{-2}) is significantly higher than the surface sulfur concentration at saturation determined previously on polycrystalline nickel using WDS (41.5 ng cm^{-2} [10], 42.0 ng cm^{-2} [17]) and radiotracer techniques (44.5 ng cm^{-2} [18]).

The NanoSIMS data (expressed in ng cm^{-2}) taken from Refs. [11,12] were also plotted in Figure 4 for direct comparison with the new AES and WDS data reported here. The methodology of GBS quantification using NanoSIMS is described in detail in Refs. [11,12], and leads to significantly lower values of GB concentration than AES and WDS. However, it should be remembered that the NanoSIMS data is obtained from a simple polished surface where any GBs can be analyzed. In contrast, the AES and WDS measurements are undertaken on a fracture surface, where it is very likely that the GBs analyzed are those with the lowest fracture strength, i.e. with the highest amount of segregated sulfur. They can be considered as "general" GBs [9,19]. On the contrary, the values obtained using

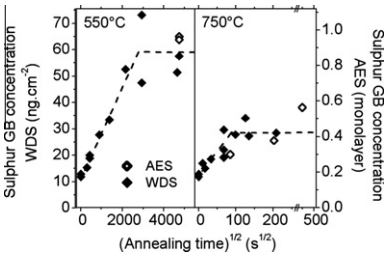


Figure 2. Kinetics of sulfur GBS in nickel at 550 and 750 °C measured by WDS and AES.

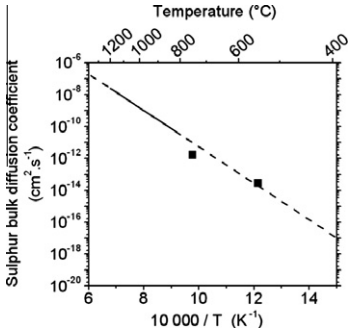


Figure 3. Arrhenius plot of the sulfur bulk diffusion coefficient in nickel. The two points are the values determined in this work at 550 and 750 °C. The solid line represents the data determined by Vladimirov et al. [7] and the dashed line is an extrapolation.

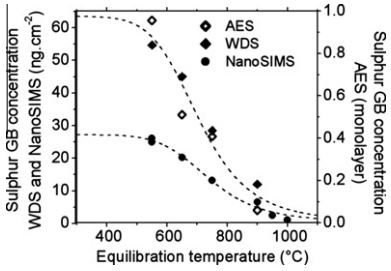


Figure 4. Temperature-dependence of sulfur GB segregation at equilibrium in nickel, measured using AES, WDS and NanoSIMS. The NanoSIMS data are from Refs. [11,12] and were obtained using the same material as in this work. The dotted lines were obtained using Eq. (1).

NanoSIMS are averaged over all types of grain boundaries, not only those with the highest GBS. Consequently, the GBs analyzed using NanoSIMS can be described as a spectrum of both general and special GBs. The discrepancy observed in Figure 4 between the results from AES/WDS and those from NanoSIMS (a factor of ~2) is quite similar to that obtained by Keast et al. [20]. In that study, the bismuth GBS determined in a copper specimen using AES was found to be significantly higher than the value measured on a thin cross-section of the same material using scanning transmission electron microscopy/energy-dispersive X-ray spectroscopy.

Fitting Eq. (1) to the NanoSIMS data gives $\Delta H = -97.1 \pm 1.5$ kJ mol^{-1} and $X_{Max} = 28.1 \pm 1.2$ ng cm^{-2} $= 0.42 \pm 0.02$ monolayer (again, the segregation entropy was neglected here). The ΔH value is very similar to that obtained using AES/WDS, but the GB sulfur concentration at saturation X_{Max} is significantly lower. Since the type of GBs investigated by AES/WDS (general GBs) and NanoSIMS (general and special GBs) is not the same, this would suggest that in the Ni–S system X_{Max} is strongly affected by GB type, but not the free energy of segregation, ΔG.

The main conclusions of this paper can be summarized as follows:

- Sulfur GBS in nickel was investigated using AES and WDS. There is a linear correlation between the two techniques showing that a monolayer in the grain boundary contains 1.3×10^{15} sulfur atoms per cm^2.
- The segregation enthalpy ΔH and the GB concentration at saturation X_{Max} were determined in the temperature range 550–900 °C to be $\Delta H = -93.9 \pm 2.5$ kJ mol^{-1} and $X_{Max} = 0.97 \pm 0.09$ monolayer $= 65.8 \pm 6.1$ ng cm^{-2}.
- The segregation values measured using AES/WDS in this work are approximately double those obtained using NanoSIMS in previous studies on the same material. This result was attributed to different GB types. As the NanoSIMS analysis is undertaken on a polished surface, any GB can be analyzed. On the contrary, using AES or WDS, only fractured GBs (i.e. those with the highest amount of GBS) can be analysed.

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